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Experiments on the effect of engine speed, load, equivalence ratio, spark timing and coolant temperature on the energy balance of a turbocharged hydrogen engine

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ABSTRACT

Internal combustion engines are used to translate the chemical energy of fuel to the brake power and over two-third of energy is wasted in the form of exergy and anergy in this process. An experimental energy balance analysis can make the engine energy flow clear and help to recycle the exergy of exhaust gas to increase brake thermal efficiency. In this paper, the effect of five different parameters on the energy distribution was investigated using heat balance test data from a 2.3 L turbocharged hydrogen engine. The results show that when proportion of exhaust gas energy ranges from 24.1% to 36.4% with various engine speeds, brake thermal efficiency can increase 3.69% at 2000 rpm and 7.67% at 4000 rpm with various load, which means that high engine speed and load are beneficial to increase exergy of total system. Both Engine speed and brake thermal efficiency increase with equivalence ratio ranging from 0.4 to 0.9, while proportion of energy of cooling system decreases, which means that both power and economic ascend with increasing equivalence ratio and anergy of total system decreases at low engine speed and load. Furthermore, proportion of exhaust gas energy can increase by 3% at 75% load and 2.5% at 50% load with spark timing closing to TDC and NOx emission is decreasing. Finally, the variation of the coolant temperature has an almost negligible effect in terms of brake thermal efficiency but it can decrease anergy of total system.

1. Introduction

Internal combustion engines are part of the powertrains for automobiles, ships and gen-set application at present. With increasingly stringent emission regulations and the fossil energy depletion, increasing brake thermal efficiency (BTE) and reducing emission are getting urgent for internal combustion engines. Exhaust gas and cooling process account for a great deal of wasted thermal energy and this wasted energy includes a lot of exergy can be recycled to increase BTE. Heat balance analysis is an efficient way to know the energy flow and then put forward a highly potential method to reduce fuel consumptions of the engines [1]. Due to the benefit of energy saving, many researchers have done much work in conventional engine [2–5]. For example, Payri et al. studied the effect of the engine speed, load, coolant temperature, inlet temperature and start of injection (SOI) on each energy distribution in an engine map of a direct injection Diesel engine [3]. Their results showed that the variation of the coolant temperature has an almost negligible effect in terms of BTE whilst cooling the air yields in an improvement about 1% and advancing the SOI about 1.5%. Martín et al. studied the effect of three parameters (low/high reactivity fuel ratio, injection timing and exhaust gas recirculation (EGR) rate) on reducing engine emissions and fuel consumption in a single-cylinder engine operating with dual-fuel [4]. Their results showed that increasing the low reactivity fuel can increase the BTE about by 1% and injection timing swept can increase by up to 4% but EGR has a limited effect.

As another way to solve the fossil fuel depletion, alternative fuel for automotive engines has been more stimulated in recent years. Heat balance analysis can not only improve the BTE for conventional engines, but also the alternative fuel engine [6–9]. For example, Yusri et al. carried out the energy balance experiment for the 2-butanol–gasoline percentage volume ratios of 5:95 (GBu5), 10:90 (GBu10) and 15:85 (GBu15) of gasoline to butanol [6]. Their results mainly exhibited an improvement in effective power, cooling energy and exhaust energy by average differences of 3.3%, 0.8% and 2.3% for GBu15 compared with G100. Das et al. have used the exergy analysis to study three compositions of biogas: BG93, BG84 and BG75 (containing 93%, 84% and 75% of CH4 by volume respectively) on a small CI engine in dual fuel mode [7]. Their results showed that biogas dual fuel could replace 80–90% of diesel fuel at lower engine loads. Under higher loads, total

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Nomenclature

PFI	port fuel injection
BTE	brake thermal efficiency
SOI	top dead center start of injection
BTDC	before top dead center
EGR	exhaust gas recirculation
WOT	wide open throttle
PMEP	pumping mean effective pressure
λ	equivalence ratio
ST	spark timing
$\dot{Q}_{H_2}$	chemical energy of fuel, kJ/kg
$\dot{M}$	mass flow, kg/h
q_{LHV,H_2}	heat value of hydrogen, MJ/kg
$\dot{M}_a$	air mass flow, kg/h
h^{sens}	specific enthalpy, kJ/kg/K
$\dot{M}_{cooling}$	mass flow of the coolant, m^3/h
$\dot{M}_{oil}$	mass flow of the oil, m^3/h
$\dot{Q}_{cooling}$	energy taken away by the coolant, kW
$\dot{Q}_{oil,b}$	energy taken away by the oil, kW
$\dot{Q}_{exh}$	energy taken away by exhaust gas, kW
ω_{NO}	mass fraction of NO in the exhaust pipe
$\dot{Q}_{inter,cooler}$	energy taken away by intercooler, kW
N	effective work, kW
c_p	specific heat capacity, kJ/kg/K
T	temperature, K
T_{tq}	torque
$\dot{Q}_{mis}$	uncounted heat loss, kW
n	engine speed, rpm
p	pressure
MBT	minimum best timing
HICE	hydrogen internal combustion engine

irreversibility of the engine was increased from 59.56% for diesel operation to 61.44%, 64.18% and 64.64% for the BG93, BG84 and BG75 biogas compositions respectively. Furthermore, combustion irreversibility was found to be decreasing with higher CO_2 concentrations in biogas. BG93 showed comparable results to that of diesel operation with 26.9% and 27.4% second-law efficiencies respectively. Li et al. studied the energy distributions of biogas using heat balance test [8]. They summarized proportion of BTE in 28.45%, exhaust gas in 40.34%, cooling system in 26.86% and heat radiation and convection from the engine only accounted for 2.99%. Gharehghani et al. investigated the thermal balance and performance of a turbocharged gas spark ignition engine at full and half loads and at different cooling fluid temperatures [9]. Results indicated that increasing the engine load and coolant temperature would lead to the increase of the percentage of transferred energy to the exhaust gas and the decrease of the percentage of coolant energy. Other alternative fuels such as alcohols [10], biodiesels [11] were also investigated using the heat balance test. By summarizing these achievements it can be found that the property of fuel property is the main reason accounting for the differences among different alternative fuel engines.

Due to the properties of hydrogen such as the faster combustion velocity, shorter quenching distance and higher RON than the other fuels [12], the energy distribution of hydrogen engine is different from the traditional and other alternative engines. With exergy analysis method, Subramanian et al. have studied the maximum available work and irreversibility (mixing, combustion, unburned, and friction) of a dual-fuel diesel engine (hydrogen-diesel) [13]. They found that the maximum available work of the diesel engine at rated load increased from 29% with conventional base mode (without H_2) to 31.7% with dual-fuel mode (18% H_2 energy share) whereas total irreversibility of the engine decreased drastically from 41.2% to 39.3%. Yu¨ksel et al. carried out the energy balance experiment in a gasoline engine using three different hydrogen mass flows as supplement [14]. Their results showed that addition hydrogen could decrease the energy of cooling water and unaccounted heat loss and increase the BTE by rate of 5%. Ozcan et al. studied the energy balance of a nature gas engine using hydrogen as supplement with a zero dimensional, two-zone computational model [15]. They found that increasing hydrogen content and lean burn have considerably affected the exergy transfers, irreversibility and second law efficiency. With the increasing hydrogen content, the irreversibility produced during combustion decreased, and the second-law efficiency sharply increased at near the lean limit. These results showed that the hydrogen as supplement can change the energy distributions for other fuel engine. Furthermore, Navale et al. carried out the energy balance experiment in the speed range of 1100 rpm to1800 rpm [16]. Results have shown that maximum brake power was reduced by 19.06% and peak BTE is increased by 3.16% in the case of hydrogen operation than the gasoline engine. Das et al. [17] introduced a 2.5 L turbocharged-intercooled hydrogen engine that can obtain a power of 66.7 kW at 3200 rpm. Results showed that BTE were above 30% at all engine speed ranges and a maximum BTE of 38% was obtained at 2000 rpm. Ford Groups studied the energy distribution in a 4.0 L six-cylinder port-injected hydrogen-fueled engine operating by natural aspiration and turbocharging [18]. They found that a BTE of 35.5% can be achieved at 1500 rpm. When increasing equivalence ratio (the ratio of the actual air-fuel ratio to the theoretical air-fuel ratio, λ), the BTE ranged from 32.8% to 34.8% and the exhaust fraction generally increased and residual decreased. These energy balance analysis results were important achievements for hydrogen engine. However, these results didn't consider different parameters (such as engine speed, load, equivalence ratio, spark timing (ST) and coolant temperature) from global heat balance and put forward reliable strategies to solve the contradiction among power, economic and emission. Thus, this paper has studied the energy balance based on different parameters and presented some controlling strategies at different working conditions using test data of a 2.3L turbocharged hydrogen engine.

2. Material and method

2.1. Theoretical method

Heat balance test can be used to explore the effects of different parameters, such as engine speed, load, equivalence ratio, spark timing, and coolant temperature, on the characteristics of different energy distributions for a turbocharged hydrogen internal combustion engine (HICE). Before performing the tests, theoretical method must be used to analyze the energy distribution. Fig. 1 shows the energy flows

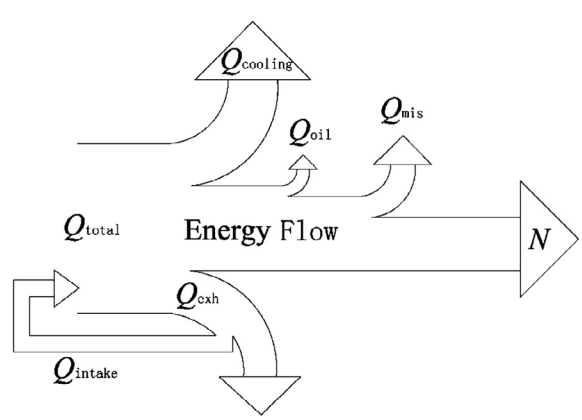

Fig. 1. Sketch of the energy flows considered.

considered of a turbocharged HICE. Based on the sketch and the first law of thermodynamics, the calculated formulas of different energy distributions can be obtained as followings:

A. Intake energy

(i) Energy of fuel

$$\dot{Q}_{H_2} = \dot{M}_{H_2} q_{LHV,H_2} \tag{1}$$

where, $\dot{Q}_{H_2}$ represents the chemical energy of fuel, kJ/kg; $\dot{M}_{H_2}$ denotes the hydrogen mass flow, kg/h; and q_{LHV,H_2} represents the heat value of hydrogen, 120 MJ/kg.

(ii) Enthalpy of intake air

$$H_a^{sens} = \dot{M}_a h_a^{sens} \tag{2}$$

where, $\dot{M}_a$ represents the air mass flow, kg/h; and h_a^{sens} represents the specific enthalpy of intake air at the intake temperature, kJ/kg/K.

(iii) Enthalpy of hydrogen

$$H_{H_2}^{sens} = \dot{M}_{H_2} h_{H_2}^{sens} \tag{3}$$

where, $h_{H_2}^{sens}$ represents the specific enthalpy of hydrogen at the intake temperature, kJ/kg/K.

B. Leaving energy

(iv) Coolant energy

$$\dot{Q}_{cooling} = \dot{M}_{cooling} c_{p,cooling} (T_{out,cooling} - T_{in,cooling}) \tag{4}$$

where, $\dot{Q}_{cooling}$ is the energy taken away by the coolant, kW; $\dot{M}_{cooling}$ is the mass flow of the coolant, m^3/h; $c_{p,cooling}$ is the specific heat capacity of the coolant, kJ/kg/K; $T_{in,cooling}$ and $T_{out,cooling}$ are the temperature of the inlet and outlet coolants, respectively, K.

(v) Energy of oil

$$\dot{Q}_{oil} = \dot{M}_{oil} c_{p,oil} (T_{out,oil} - T_{in,oil}) \tag{5}$$

where, $\dot{Q}_{oil,b}$ is the energy taken away by the oil, kW; $\dot{M}_{oil}$ is the mass flow of the oil, m^3/h; $c_{p,oil}$ is the specific heat capacity of the oil, kJ/kg/K; $T_{in,oil}$ and $T_{out,oil}$ are the temperature of the inlet and outlet oils, respectively, K.

(vi) Energy of exhaust gas

For hydrogen engine, exhaust gas mainly includes H_2O, N_2, O_2 and NOx. The concentration of HC and CO is extremely small because they both result from the combustion of lubricant. NOx mainly consists of NO. Thus the formula of exhaust gas energy can be calculated as followings:

$$\dot{Q}_{exh} = (\dot{M}_{N_2} c_{p,N_2} + \dot{M}_{O_2} c_{p,O_2} + \dot{M}_{H_2O} c_{p,H_2O})(T_{exh} - T_a) + \dot{M}_{H_2} + \dot{M}_a \omega_{NO} c_{p,NO} \tag{6}$$

where, $\dot{Q}_{exh}$ is the energy taken away by the exhaust gas, kW; $\dot{M}_{N_2}$, $\dot{M}_{O_2}$ and $\dot{M}_{H_2O}$ represent the mass flow of N_2, O_2 and H_2O in the exhaust pipe respectively, kg/h; c_{p,N_2}, c_{p,O_2}, c_{p,H_2O} and $c_{p,NO}$ represent the specific heat capacity of the N_2, O_2 and H_2O respectively, kJ/kg/K; and ω_{NO} represents the mass fraction of NO in the exhaust pipe.

(vii) Energy of the turbocharger intercooler

$$\dot{Q}_{inter.cooler} = \dot{M}_a c_{p,a} (T_{out,inter.cooler} - T_{in,inter.cooler}) \tag{7}$$

where, $\dot{Q}_{inter.cooler}$ is the energy taken away by the turbocharger intercooler, kW; $c_{p,a}$ is the specific heat capacity of the air at the outlet of the compressor, kJ/kg/K; and $T_{in,inter.cooler}$ and $T_{out,inter.cooler}$ represent the temperature of the inlet and outlet air through the intercooler, respectively, K.

(viii) Effective work

$$N = 2 \times \pi \times n \times T_{tq} \times 10^{-3} \tag{8}$$

where, N represents the effective work, kW; n denotes the engine speed; and T_{tq} represents the torque.

(ix) Uncounted heat loss

Uncounted energy is a general term for a series of energies; $\dot{Q}_{mis}$ represents this energy, which mainly includes the following four parts: (1) the convective heat transfer and heat radiation between the cylinder wall and head, turbocharger system, cooling system and intake and exhaust pipes with the ambient; (2) the leakage of the hydrogen and in-cylinder gas; (3) the loss between the experimental non-equilibrium and equilibrium state; and (4) the error caused by the combustion of lubricating oil.

The energy balance formula of a turbocharged hydrogen engine can generally be expressed as follows:

$$\dot{Q}_{H_2} + \dot{M}_a h_a^{sens} + \dot{M}_{H_2} h_a^{sens} = N + \dot{Q}_{cooling} + \dot{Q}_{oil} + \dot{Q}_{exh} + \dot{Q}_{inter.cooler} + \dot{Q}_{mis} \tag{9}$$

2.2. Test process

Tests were performed in a 2.3 L, four-stoke, port fuel injection (PFI) hydrogen internal combustion engine with a turbocharger. The main parameters of the hydrogen engine are shown in Table 1 and the sketch of the bench is shown in Fig. 2. Table 2 shows the uncertainty and range of the sensors.

In this study, the heat balance test was performed with FC200 test system, which mainly included a CW250 eddy current dynamometer, cooling system for oil and coolant, turbocharger intercooler system and a data acquisition system. The dynamometer is used to measure power and motor. The cooling system is connected to the engine's oil and coolant system, the temperatures of which were controlled through convective heat transfer with the cooling water in water tower as shown in Fig. 2. The temperatures of the oil and coolant and the mass flow are measured by Pt100 and turbine flow meters. Given that the variation of the temperature at low engine speed and low load was extremely small, the accuracy of the Pt100 must be below 1 °C, as shown in Table 2. All of the temperatures and flow meters can be used to analyze the cooling water and oil energy distribution. The turbocharger intercooler system was connected to the compressor outlet and was used to control the intake air temperature through convective heat transfer with the cooling water. The system also measured the temperature and pressure of intake air, which can be used to analyze the characteristic of energy

Table 1
Parameters of a 2.3 L hydrogen engine with a turbocharger.

Engine type	Hydrogen engine
Bore	90 mm
Stroke	90 mm
Cylinders	4
Crank radius	45 mm
Connecting rod radius	153 mm
Compression ratio	9.3
Exhaust valve open	53.64°BBDC
Exhaust valve close	44.36°ATDC
Inlet valve open	23.3°BTDC
Inlet valve close	58.7°ABDC
Maximum torque @ 3000 rpm	161 N m
Maximum power @ 4000 rpm	59.4 kW

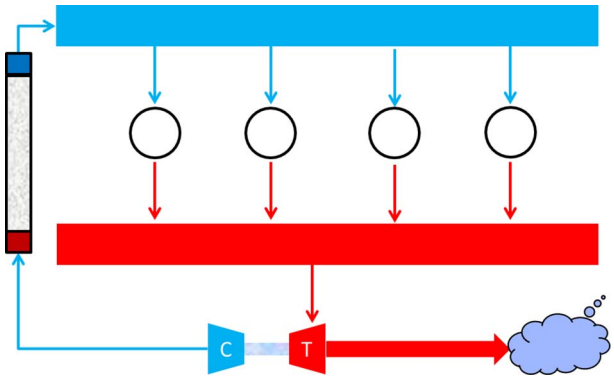

Fig. 2. Diagram of the turbocharged hydrogen internal combustion engine.

Table 2
Average uncertainty of the main measured parameters.

Parameter	Uncertainty	Range
Engine speed	± 1	0–8000 rpm
Torque	± 0.4%FS[a]	0–1100 N m
Pt100	± 0.5	0–423 K
Thermocouple K type	± 1	0–1473 K
Intake pressure	± 0.4%FS[a]	-0.15–0.25 MPa
Exhaust pressure	± 0.4%FS[a]	0–0.1 MPa
Oil mass flow rate	± 0.5%FS[a]	0.6–6 m^3/h
Cooling mass flow rate	± 0.5%FS[a]	1.5–15 m^3/h
Air mass flow rate	± 1% FS[a]	0–450 kg/h
Hydrogen mass flow rate	± 0.1% FS[a]	0–6.5 kg/h

[a] FS: Full scale.

distribution. The temperature of the exhaust gas is different at different position. To reduce the error, four exhaust temperature sensors were installed on the manifolds of the four cylinders and two sensors were installed before and after the turbine to measure the average temperature of the exhaust gas. AVL DiGas 4000 Light was also installed on the exhaust pipe to measure the concentration of different exhaust gases such as NOx, CO_2 and HC. The specific heat capacities of exhaust gas were acquired by interpolation calculation according to the measured data. The air mass flow was measured by a hot line air mass flow meter. The hydrogen mass flow was measured by two flow meters (i.e., CMF010 and CMF025). With these data and the Eq. (6), the exhaust gas energy distribution can be analyzed. Several other parameters, such as the cylinder pressure can be used for the measurement to obtain a credible map. In this paper, the equivalence ratio of λ means that the ratio of the actual air-fuel ratio to the theoretical air-fuel ratio as the following:

$$\lambda = \frac{\text{the actual fuel}-\text{air ratio}}{\text{the theory fuel}-\text{air ratio}} \tag{10}$$

Test points to study the effect of different parameters on the energy distributions can be divided into four parts as shown in Table 3. Although a complete measurement matrix is available for the parametric studies, only the results from 1500 rpm to 4000 rpm with interval of 500 rpm and parts of loads were presented for the sake of brevity. These operating points are selected, because these engine speeds allowed for the assessment of the effect of reducing the heat transfer time by one half, while the same trends were observed in all the operating points measured in this study.

3. Results and discussions

3.1. Engine speed and load variation

According to the theory in Section 2.1, engine speed and load are the two most important factors that affect energy distribution. First, the effect of engine speed is as shown in Fig. 3. Several conclusions can be obtained from the Fig. 3.

(1) BTE is not sensitive to the engine speed; however, the effective work increases with engine speed. According to Otto cycle theory, the isochoric process of Otto cycle has effect on the BTE and this process is affected by the combustion velocity. The engine speed can increase the turbulence intensity, which can increase the combustion velocity but the influence is very small for hydrogen engine. The combustion velocity of hydrogen-air mixture is mainly affected by the equivalence ratio. Thus, the engine speed has effected on the BTE but the value is very small. The increase of effective work with engine speed can be explained by the air and hydrogen mass, as shown in Fig. 4. Air and hydrogen mass increase with engine speed because the increase of piston velocity leads to an increase in cycle numbers per unit time. The result is that more fuel energy will release more effective work.
(2) The energy taken away by the cooling system is a significant part of the total energy of hydrogen, which reflects the heat transfer process between the high-temperature cylinder gas and the cooling system. Given the short quenching distance of hydrogen and air (only 0.64 mm), the thermal boundary layer between the gas and the cylinder wall is extremely thick, and the heat transfer is different from the traditional engines. As shown in Fig. 3, the energy taken away by the cooling system can be divided into two parts, namely, oil and coolant. The energy taken away by two types of cooling medium increases with engine speed; however, the proportion to the total energy changes minimally. The reason can be explained by two main factors, namely, heat transfer time and in-cylinder gas temperature. Heat transfer time decreases with the increase of engine speed, which leads to the decrease of heat transfer loss between the cooling system and the in-cylinder gas. Meanwhile, the combustion temperature in the cylinder ascends with the engine speed, which means that the heat transfer loss

Table 3
Measurement set.

Title	Range	Step
(1) Engine speed and load variations		
Speed	1500–4000 rpm	500 rpm
Load	Minimum-wide open throttle (WOT)	2 deg
$T_{cooling}$	80 °C	–
T_{oil}	90 °C	–
λ	0.55	–
ST	Maximum brake torque (MBT)	–
(2) Coolant temperature variations		
Speed	2500 rpm	–
Load	Minimum-WOT	2 deg
$T_{cooling}$	60 °C − 80 °C	10 °C
T_{oil}	90 °C	–
λ	0.55	–
ST	MBT	–
(3) Equivalence ratio variations		
Speed	2500 rpm	–
Load	25%	–
$T_{cooling}$	80 °C	–
T_{oil}	90 °C	–
λ	0.4–1.1	0.05
ST	MBT	–
(4) IT variations		
Speed	2500 rpm	–
Load	50%, 75%	–
$T_{cooling}$	80 °C	–
T_{oil}	90 °C	–
λ	0.55	–
ST	− 5–25 ° CA BTDC	5° CA

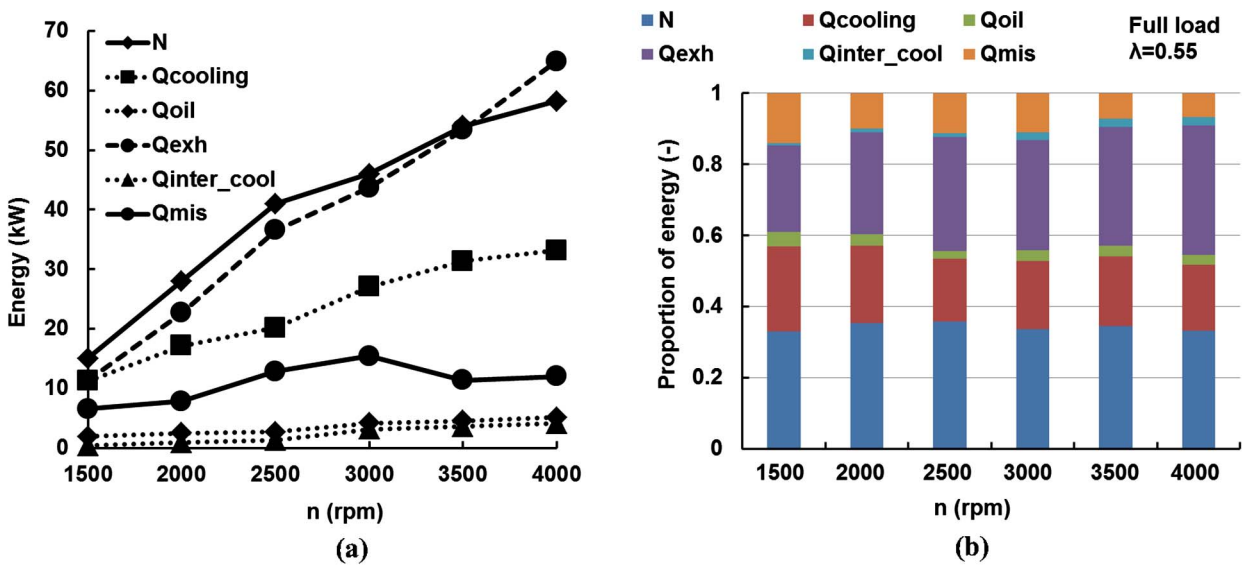

Fig. 3. Thermal balance at full load and different engine speeds.

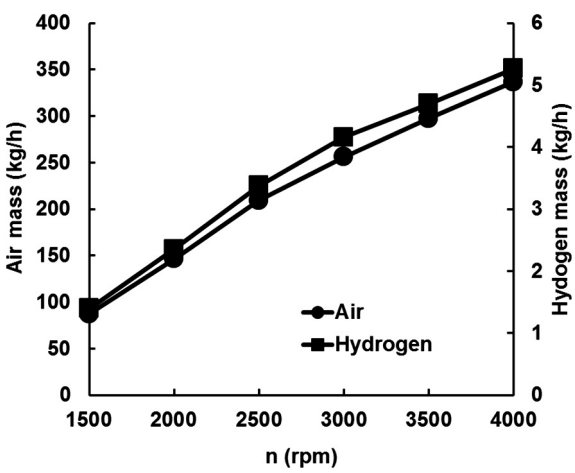

Fig. 4. Air and hydrogen mass change with the engine speed.

increases because the temperature difference between the in-cylinder gas and the cooling system is increasing. These two factors result in the increase of energy with minimal varying proportion.

(3) The energy of exhaust gas accounts for about one-third of total energy, and this energy includes high proportion of exergy, which can be used to increase BTE. As shown in Fig. 3, the energy of the exhaust gas proportion increases with the engine speed. Based on Eq. (6), the main factors that affect the exhaust gas energy are the mass and temperature of exhaust gas. The exhaust gas mass increases with the engine speed, as shown in Fig. 4; the temperature also increases with engine speed, as shown in Fig. 5. The result of the two increasing factors is that the energy of exhaust gas increases with engine speed. The two factors also mainly result in the proportion of exhaust gas increasing with engine speed. Another reason for the proportion is that the turbocharged pressure at the outlet of compressor should be controlled to avoid knock, which leads to part of the exhaust gas being released through the waste gate at high engine speed. As a result, the effective work at high engine speed decreases and the energy of exhaust gas increases.

(4) The proportion of uncounted energy decreases with the increase of engine speed, as shown in Fig. 3. Summarizing the other energy part, including effective work, cooling system, and exhaust gas, the sum of the three proportions increases with the engine speed; thus, the proportion of uncounted energy decreases, according to Eq. (9).

Fig. 6 shows the distribution of different energy parts at 2000 rpm and 4000 rpm, MBT, $\lambda = 0.55$. With the throttle angle changing from the minimum to WOT, effective work increases from 15 k W to 28 kW, and the energy taken away by exhaust gas increases from 13 kW to 23 kW at 2000 rpm. The energy taken away by the cooling system ranges from 16 kW to 19 kW. Three types of energy are analyzed. First, the mass of air and hydrogen increases the loads, which results in the increase of total energy, thereby increasing the effective work and the energy of exhaust gas. The exhaust gas temperature also increases, which can also increase the energy of exhaust gas. Two reasons can explain the phenomenon of the cooling system. First, the temperature of the in-cylinder gas changes minimally with the load when the equivalence ratio is kept constant. Second, the heat transfer time remains constant at the same engine speed. These two factors result in the heat transfer loss between the in-cylinder gas and the cooling system changing minimally. The effective work increases from 27 kW to 59 kW, and the energy taken away by the exhaust gas increases from 35 kW to 65 kW at 4000 rpm. The energy of the exhaust gas is larger than effective work at 4000 rpm, which is different from that at 2000 rpm. This occurrence is caused by the absorption of energy of the exhaust gas by the turbine to boost the pressure of inlet air, which increases the pressure and temperature at the end of the combustion. Accordingly, the average temperature in the cylinder is higher than the temperature at 2000 rpm, and the exhaust gas temperature is also higher at

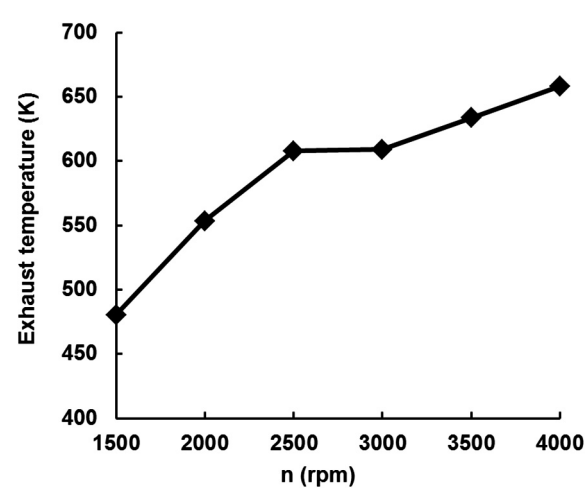

Fig. 5. Exhaust temperature changes with the engine speed.

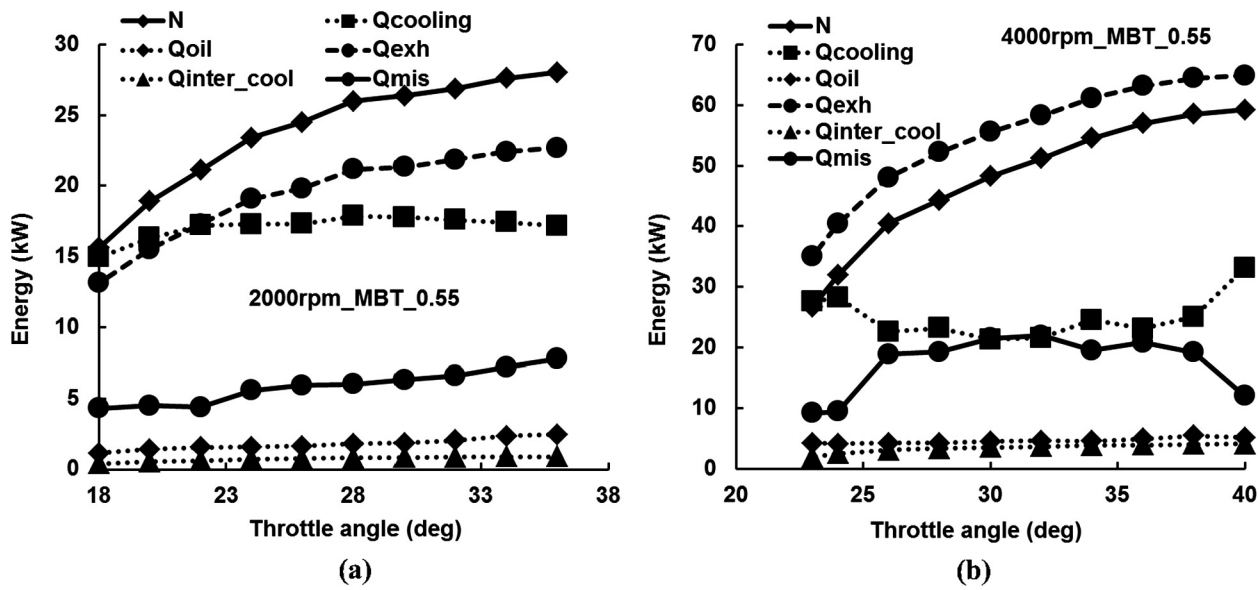

Fig. 6. Different energies change with throttle angle at different engine speeds.

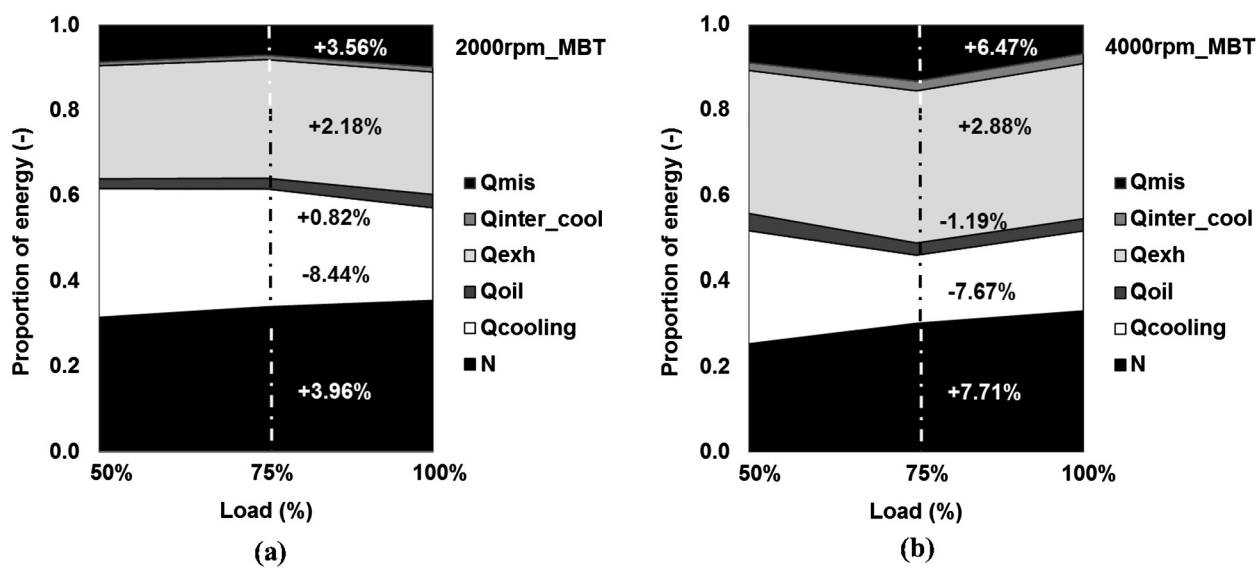

Fig. 7. Proportion of different energies change with throttle angle at different engine speeds.

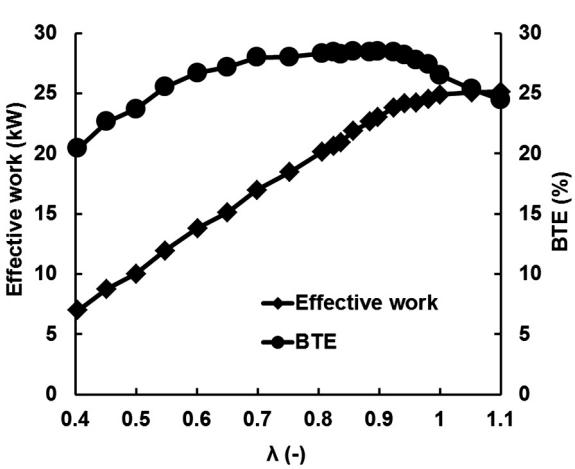

Fig. 8. Effective work and BTE change with λ.

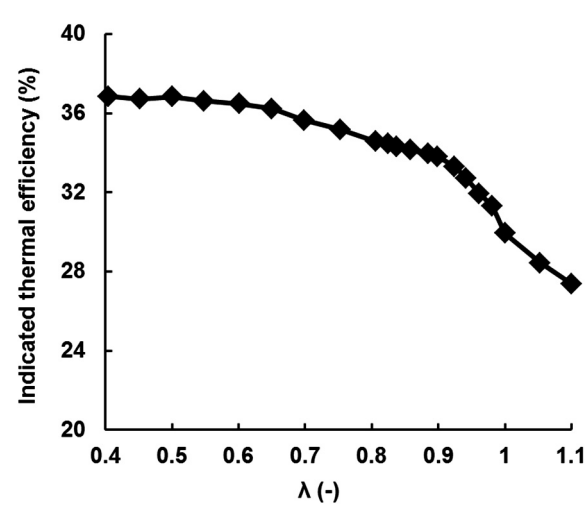

Fig. 9. Indicated thermal efficiency changes with λ.

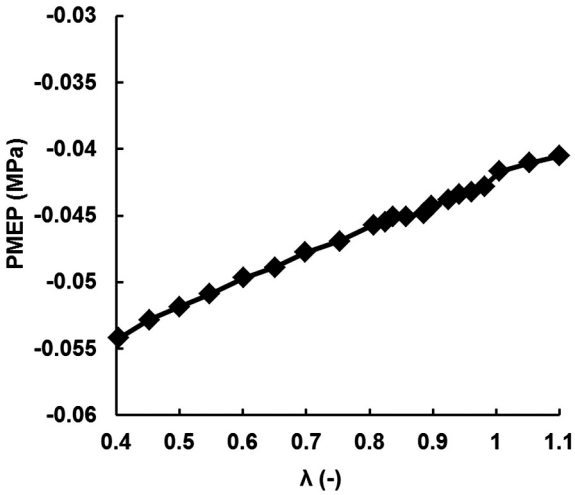

Fig. 10. Pumping Mean Effective Pressure (PMEP) changes with λ.

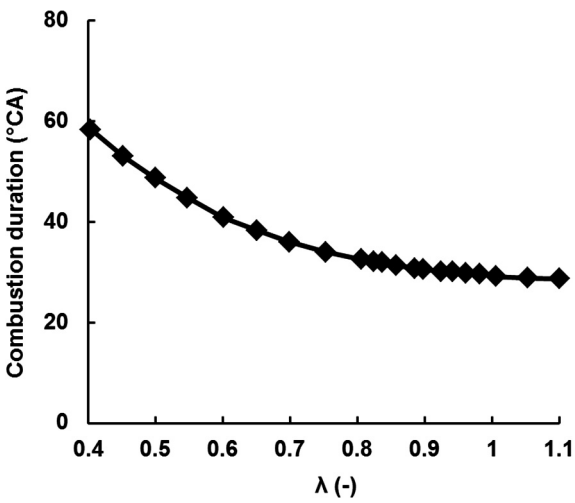

Fig. 11. combustion duration changes with λ.

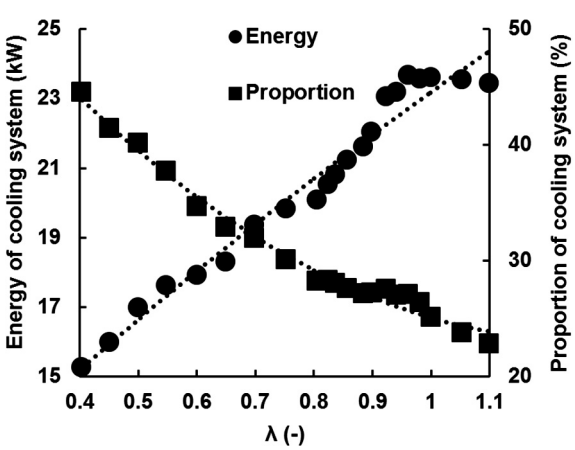

Fig. 12. Energy and proportion of cooling system changes with λ.

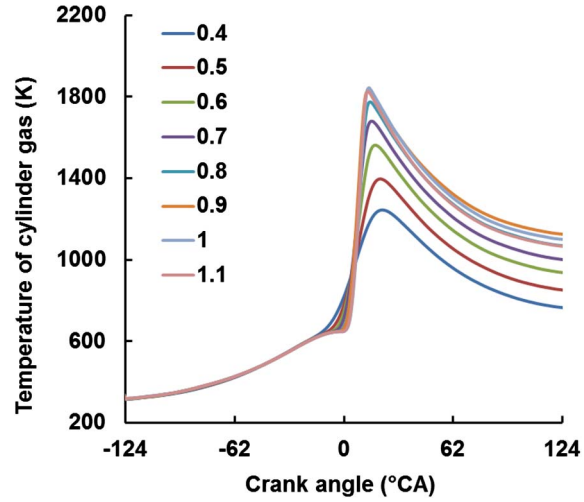

Fig. 13. In-cylinder combustion temperature changes with λ.

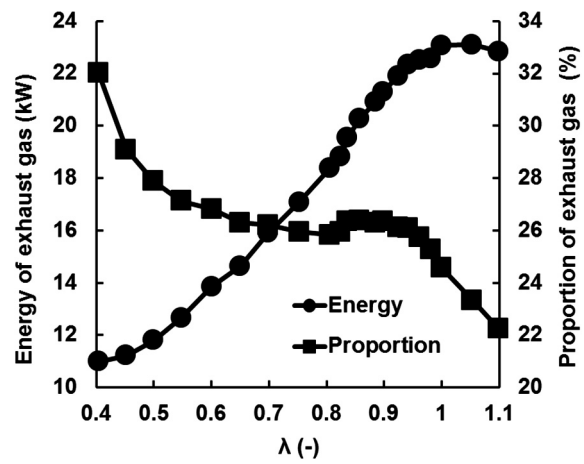

Fig. 14. Energy and proportion of exhaust gas change with λ.

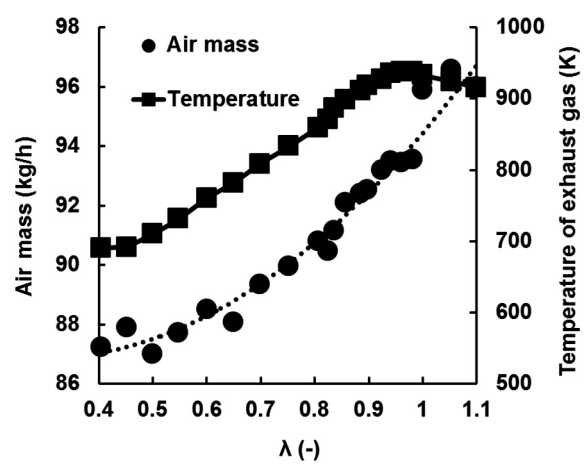

Fig. 15. Air mass and temperature of exhaust gas change with λ.

4000 rpm. This phenomenon is according to the data in Fig. 3. The trend of the energy taken away by the cooling system at 4000 rpm is similar to the one at 2000 rpm, and the energy is larger only because of the abundance of total energy.

To analyze the effect of loads on the energy distributions, two engine speeds (i.e., 2000 rpm and 4000 rpm) and three load points (i.e., 100%, 75%, and 50%) are selected, as shown in Fig. 7. BTE increases by 3.69% at 2000 rpm and 7.67% at 4000 rpm when the throttle angle ranges from the minimum to WOT. Therefore, BTE is extremely sensitive to engine load, especially at high engine speed. Two reasons can explain this phenomenon. First, throttling and pumping loss in the process of gas exchange decrease, and the fraction loss changes minimally with the increase of load when the engine speed and equivalence ratio are kept constant. Second, the pressure of inlet air increases with the load due to the effect of the turbine, thereby further reducing the

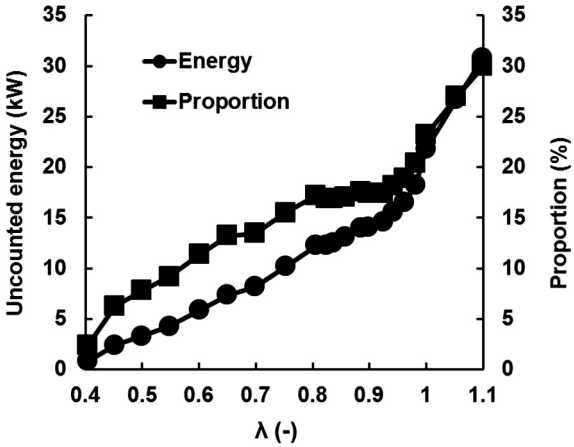

Fig. 16. Uncounted energy and proportion change with λ.

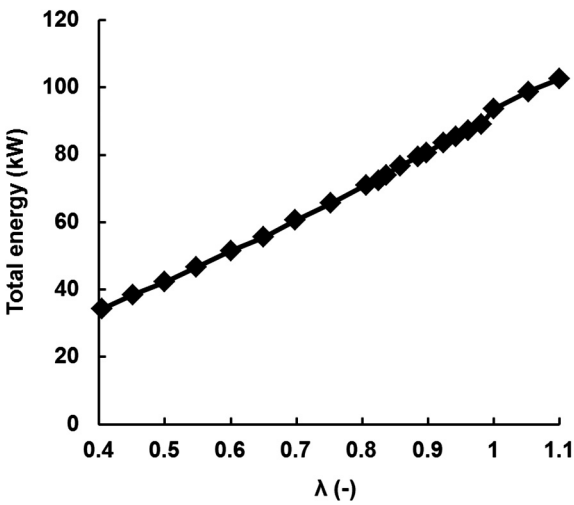

Fig. 17. Total energy changes with λ.

pumping loss. Two reasons lead to the increase of BTE with the increasing load. The proportion of exhaust gas increases by 2.18% at 2000 rpm and 2.88% at 4000 rpm; thus, the proportion of exhaust gas is not sensitive to engine load. Unlike the effect of engine speed, the proportion of the energy taken away by the cooling system is overly sensitive to the load, and it decreases by 7.62% at 2000 rpm and 8.86% at 4000 rpm. The main reason is that a low BTE will lead to a relatively high fraction loss at low load and the heat generated by the fraction will be taken away by the cooling system, which will make the proportion increase with engine load.

3.2. Equivalence ratio variation

Equivalence ratio is a considerably significant parameter for the turbocharged HICE because it can influence the combustion velocity and in-cylinder temperature. According to the analysis in Section 3.1, the in-cylinder temperature and combustion velocity influence not only the effective work and BTE but also the energy of the exhaust gas, the heat transfer of the cooling system, the energy absorbed by the turbine, and the work of the compressor. This section studies the effect of the equivalence ratio on the characteristic of energy distribution using a test at 2500 rpm, throttle angle of 17°, MBT, and λ = 0.4–1.1 interval of 0.05. This operating condition is selected for two reasons. First, the turbine cannot absorb sufficient exhaust gas energy to increase the inlet pressure; thus, the effective work is the most critical problem among the contradiction of power, economy, and emission at low engine speed and load. The equivalence ratio can increase the effective work. Thus, the test data are fit for these operating conditions. Second, the mass of the hydrogen nozzle is limited by the injection pressure, which cannot exceed 5 bars. At high engine speed and load, the increasing equivalence ratio requires that the pulse width of the sprayed hydrogen should be accordingly increased, which may lead to backfire.

3.2.1. Effective work

As shown in Fig. 8, effective work when λ = 1.0 is 4 times of that when λ = 0.4 at the same engine speed and throttle angle, which proves that the equivalence ratio can increase the power of the hydrogen engine. Analysis of the trend of BTE changing with equivalence ratio reveals that BTE increases with equivalence ratio; however, the increased velocity decreases, especially when λ > 0.9. According to Otto cycle theory and lean combustion, a low equivalence ratio will lead to a high BTE. However, BTE in Fig. 8 has several differences with the theory. The discrepancy can be explained from the following aspects. First, the theory cycle of BTE pertains to indicated thermal efficiency. As shown in Fig. 9, the indicated thermal efficiency ranges from 36.7% to 29.9% when the equivalence ratio changes from 0.4 to 1.0. Therefore, the indicated thermal efficiency for the operating conditions at 2500 rpm is based on Otto cycle theory and lean combustion. Second, BTE is equal to the indicated thermal efficiency minus the mechanical

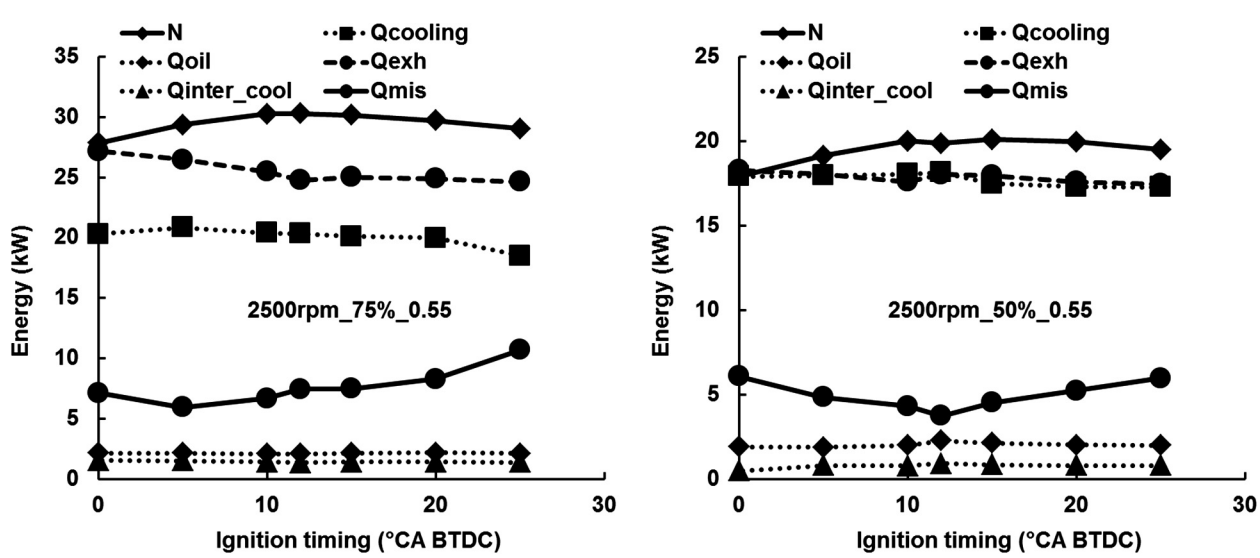

Fig. 18. Different part energies change with ST at different loads.

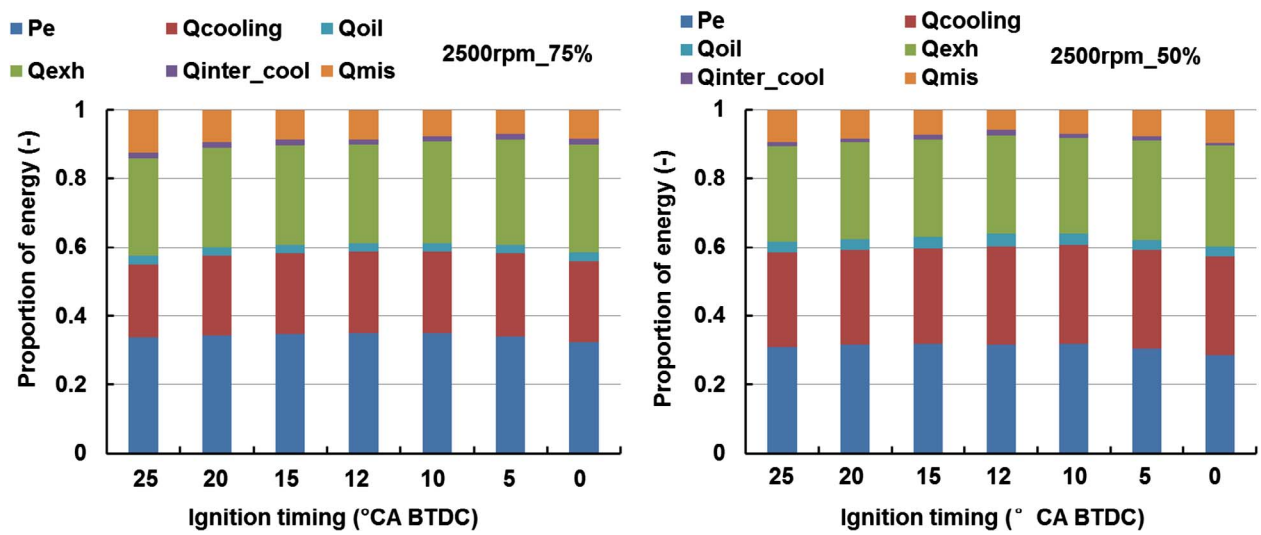

Fig. 19. Proportion of different energies change with ST at different loads.

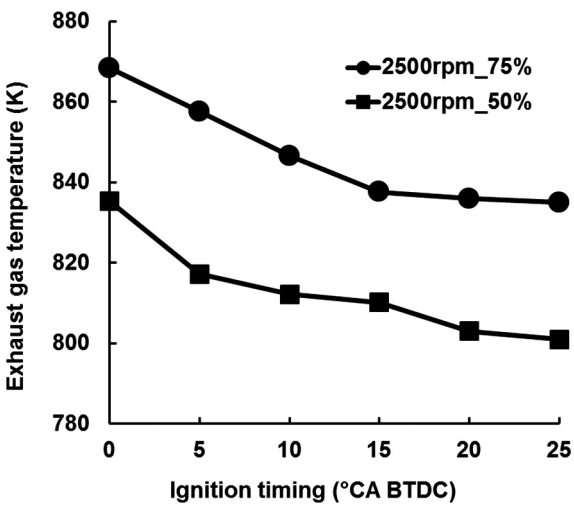

Fig. 20. Exhaust gas temperature changes with IT at different loads.

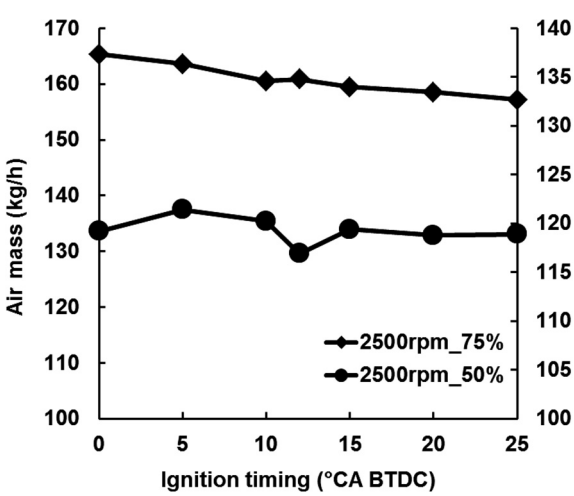

Fig. 21. Air mass changes with IT at different loads.

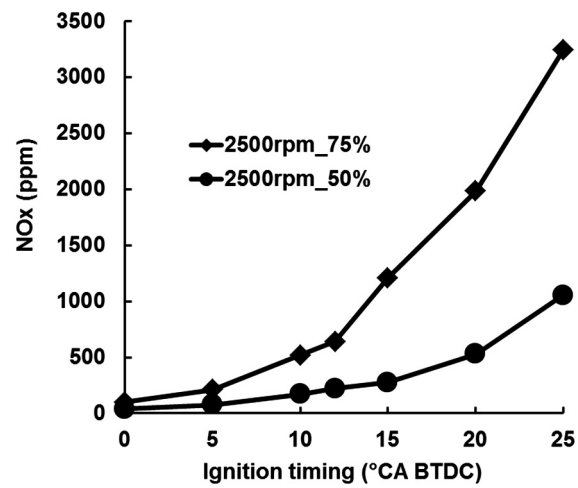

Fig. 22. NOx emission changes with ST at different loads.

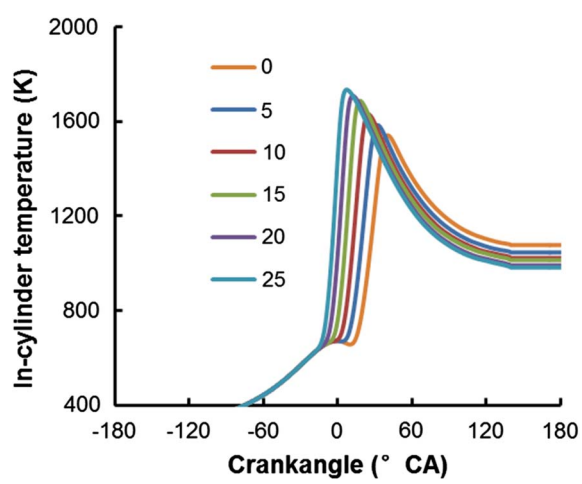

Fig. 23. In-cylinder temperature changes with ST.

loss. The mechanical loss has two main parts: pumping loss and friction loss. The friction loss changes very small at the same engine speed and throttle angle. But pumping loss decreases from 0.054 MPa to 0.041 MPa with the increase of equivalence ratio, as shown in Fig. 10.

Third, combustion speed is sensitive to and increases with the equivalence ratio. The combustion duration (0–100%) of $\lambda = 1.0$ is 29.1° CA and that of $\lambda = 0.4$ is 58.3° CA, as shown in Fig. 11. According to Otto cycle theory, a fast combustion speed indicates a long constant volume, which will make BTE high. The result of these three factors is that the

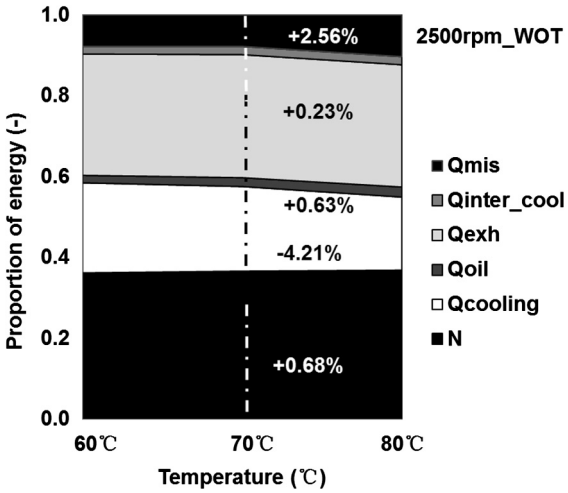

Fig. 24. Proportion of different energies change with different coolant temperatures.

BTE of the hydrogen engine increases with the equivalence ratio at low engine speed and load. Therefore, the increase of equivalence ratio not only increases effective work but also improves BTE at low engine speed and load. The controlling strategy for turbocharged hydrogen engines must be guided. When the equivalence ratio is bigger than 1.0, the BTE and effective work are decreasing.

3.2.2. Energy of cooling system

According to the analysis in Section 3.1, factors that influence the energy of the cooling system are heat transfer time and temperature difference. When engine speed and throttle angle are kept constant, the heat transfer time also changes minimally, and the temperature difference significantly influences the energy distribution for different equivalence ratios.

The trend of the cooling system is similar to those of the effective work, and the energy increases from 15 kW to 24 kW with $\lambda = 0.4–1.0$, as shown in Fig. 12. Fig. 13 can explain this phenomenon. The in-cylinder combustion temperature increases with the equivalence ratio, and the highest value can reach 1800 K at $\lambda = 1.0$. A high combustion temperature indicates a large temperature difference between the in-cylinder gas and the cooling system. As a result, the heat transfer loss increases. However, the proportion of the cooling system decreases with increasing equivalence ratio from 45% to 25%. The comparison between Figs. 12 and 8 shows that the increment of the cooling system energy is 9 kW with $\lambda = 0.4–1.0$, which is smaller than the increment of effective work (18 kW). Thus, the energy of the cooling system increases, whereas the proportion decreases with various equivalence ratios. The energy cooling system change very small with $\lambda > 1$ but the proportion is decreasing. The reason is that the total energy is increasing but energy of cooling system changes very small. The result is the proportion decrease with $\lambda > 1$. This is also the reason of exhaust gas proportion as shown in Fig. 14.

3.2.3. Energy of exhaust gas

Fig. 14 shows the trend of energy and the proportion of exhaust gas changing with the equivalence ratio, and two trends show opposite characteristics. First, the trend of the energy of exhaust gas is analyzed. As shown in Fig. 15, the air mass and temperature of the exhaust gas increase with the equivalence ratio. In Fig. 13, a high in-cylinder combustion temperature results in a high temperature of exhaust gas. The in-cylinder temperature profiles are obtained by calculating the cylinder pressure using the ideal gas equation of state. The intake manifold temperature is approximately the same as the cylinder temperature when the intake valve is closed. Utilizing this temperature and the cylinder pressure can calculate the cylinder temperature. As the equivalence ratio increases, high temperature drives the turbine to absorb the energy of the exhaust gas, which can be used to increase the pressure of the inlet air. As a result, the air mass flow rate increases from 90 kg/h to 96 kg/h with $\lambda = 0.4–1.0$. These two factors increase the energy of exhaust gas. However, the proportion of the exhaust gas to the total energy decreases with the increase of equivalence ratio. This trend is similar to that of the cooling system proportion. As the equivalence ratio changes from 0.4 to 1.0, the energy of the exhaust gas increases from 10.9 kW to 23.1 kW, but difference is smaller than that of the cooling system. Furthermore, part of the exhaust gas energy is used to increase effective work through turbine recovery, which is another reason for the trend of the proportion.

3.2.4. Energy of uncounted

Aside from the above analysis of the three parts of the energy, the distribution of uncounted energy is also shown in Fig. 16. First, uncounted energy also increases with the equivalence ratio and ranges from 0.8 kW to 24.1 kW with $\lambda = 0.4–1.0$. The total energy is shown in Fig. 17, which increases linearly with the equivalence ratio. The characteristics of effective work, energy of the cooling system, and the exhaust gas show that the uncounted energy also increases with increasing equivalence ratio. Furthermore, the proportion of uncounted energy ranges from 4% to 20% with $\lambda = 0.4–1.0$. It can be explained that the convective heat transfer and heat radiation increase with increasing equivalence ratio and the leakage of the hydrogen also increases. When the equivalence ratio is bigger than 1.0, the uncounted energy and proportion will increase drastically because the extra hydrogen will enter the exhaust gas.

3.3. Spark timing variation

Ignition timing influences not only the effective work and NOx emission but also the energy of exhaust gas through the exhaust gas temperature. Thus, studying the effect of spark timing on the energy distribution can help to solve the contradiction among power, economy, and emission of hydrogen engines.

Given the possibility of backfire at WOT, the test was conducted at 2500 rpm, $\lambda = 0.55$, and 75% and 50% of engine load. The following characteristics of energy distribution can be obtained from Figs. 18 and 19.

(1) The effect of spark timing on the effective work can be divided into two parts. First, when the spark timing is earlier than 12° CA BTEDC (MBT), effective work increases. This occurrence is caused by the increase in negative in-cylinder pressure by the early spark when the piston rises from BDC to TDC. The negative pressure increases the negative work, which leads to the decrease of effective work. Spark timing later than MBT will increase the combustion duration and decrease the constant volume. These two factors decrease the effective work and BTE.
(2) With the spark timing becoming close to the TDC, the proportion and energy of exhaust energy increase, especially at high load. It can be explained from two aspects. First, late spark timing indicates a late combustion end angle; thus, the combustion duration is close to the stroke of exhaust. Hence, the exhaust gas temperature increases with spark timing becoming close to TDC, as shown in Fig. 20. Second, the trend of air mass flow is increasing with the spark timing becoming close to TDC, as shown in Fig. 21, because the part of exhaust gas energy can be absorbed by the turbine to increase the pressure of inlet air, especially at high engine load. Thus, according to Eq. (6), the energy and proportion increases with the spark timing becoming close to TDC.(see Fig. 22)
(3) The energy and proportion of the cooling system change minimally with the change in spark timing because the spark timing hardly has an effect on the heat transfer time and temperature difference. The proportion of the cooling system at 75% load is smaller than that at

50% load, which agrees with the conclusions in Section 3.1.

(4) NOx changes from 40 ppm to 3239 ppm with the advance of spark timing from 0 to 25° CA BTDC. Factors of effect on the NOx emission have three: high combustion temperature, enough oxygen and long reaction time [19]. With spark timing ranging from 25 to 0° CA, the combustion duration is increasing, which means the reaction time is increasing. With the advance of spark timing (from 0 to 250° CA BTDC), the combustion temperature also increases significantly, leading to the increase of NOx as shown in Fig. 23. At the same time, the oxygen concentration doesn't change because the equivalence ratio keeps constant with variation of ST. Thus, the NOx emission is increasing with spark timing.

3.4. Coolant temperature variation

According to the analysis in Section 3.1, the effect of coolant temperature on the energy distribution of the cooling system is greater than the effect of oil temperature. Furthermore, the energy taken away by the cooling system is anergy, and this energy part cannot be recycled to increase effective work and BTE. Thus, studying the effect of coolant temperature on the energy distribution can help reduce the proportion of anergy and increase the proportion of exergy.

The coolant temperature is set to 60 °C, 70 °C, and 80 °C at 2500 rpm, $\lambda = 0.55$, MBT, and different throttle angles (from minimum to WOT). Fig. 24 shows the results of the variation in coolant temperature. Coolant temperature variation directly influences wall temperatures, which leads to a diminution in the temperature difference between the in-cylinder gas in the chamber and the walls, resulting in a reduction of 4.21% at 2500 rpm in the energy of the cooling system. Meanwhile, the increase in the engine wall temperature leads to an increase of 2.56% in uncounted energy, mainly due to the high heat radiation to the ambient. The effect of coolant temperature variations on N is small in both operating conditions; the brake efficiency variation is lower than 0.7% in all operating conditions. The result of exhaust gas energy is similar to that of effective work, whose variation is lower than 0.3%.

4. Conclusions

Effect of different parameters including engine speed, load, equivalence ratio, ST and coolant temperature on the energy distributions has been investigated with turbocharged hydrogen engines through heat balance test in this paper. Conclusions can be found as following:

(1) Increasing engine speed can obviously ascend all part energies and the increments of exhaust gas energy is the biggest. In addition, BTE and proportion of cooling system energy change extremely small with engine speed but the proportion of exhaust gas energy can range from 24.1% to 36.4%. Thus, when the target power is same, choosing high engine speed is beneficial to increase BTE because the exergy of exhaust gas is higher.

(2) The variation of engine load presented the greatest influence on the BTE of increasing 3.69% at 2000 rpm and 7.67% at 4000 rpm and proportion of cooling system of decreasing 7.62% at 2000 rpm and 8.86% at 4000 rpm. The proportion of exhaust gas energy increases 2.18% at 2000 rpm and 2.88% at 4000 rpm, which is lower than the other proportions. It means that engine load can not only increase BTE, but also decrease the anergy of total system. Thus, high engine speed and engine load are better choices for turbocharged HICE.

(3) Analysis of the four parts of energy distribution with equivalence ratio shows that the four parts of energy all increase with $\lambda = 0.4$–0.9; however, the increments vary. Increasing the equivalence ratio can increase effective work and BTE, thereby improving power and economy. In addition, the proportion of the cooling system energy changing minimally means that the anergy does not increase in the total system. Finally, the decrease in the proportion of exhaust gas energy is due to the energy part being absorbed by the turbine to increase the effective work; thus, increasing the equivalence ratio can improve the efficiency of the turbocharger system.

(4) Effective work and BTE show the same trend with spark timing variation and the biggest value presents at the point of MBT (12° CA BTDC). Both proportion and energy of exhaust gas both are increasing but the cooling system energy and proportion change extremely small with spark timing closing to TDC. Furthermore, considering the relationship of power, economic and NOx emission, the controlling strategy can decrease the NOx emission prior through adjusting spark timing near to TDC and then increase the effective work and BTE through the turbine absorbing the exhaust gas energy at part engine speed and loads.

(5) The variation of coolant temperature shows an almost negligible effect on the BTE (lower than 0.7%) and proportion of exhaust gas energy (lower than 0.3%). However, lower coolant temperature promotes the energy from the in-cylinder gas to the coolant and the heat transfer to the oil and the proportion of uncounted energy is decreased.

Acknowledgement

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